

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8CJ407 / MAT8MN407			
Course Title	NUMBER THEORY			
Type of Course	Major			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	4	-	60
Pre-requisites	Basic algebra of integers, basic set theory, basic proof techniques.			
Course Summary	This is a more advanced course than MAT6CJ305 / MAT8MN305 Elementary Number Theory. Here we focus on arithmetical functions, their averages, distribution of prime numbers, quadratic reciprocity and in the last open-ended section, Cryptography. Arithmetical functions are geared towards the study of prime numbers and their distribution. We provide a rigorous examination of several of them such as Mobius function, Euler's totient function, and compositions through techniques such as Dirichlet multiplication and convolution. Next we study their asymptotic behaviour using techniques such elementary estimates, partial summation and Dirichlet products. Next, we study the distribution of prime numbers. The prime number theorem is stated along with some equivalent versions and a build-up to it. Next the concept of quadratic residues, quadratic reciprocity and how to compute the same, along with applications, are studied. The open-ended part is Cryptography.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and analyse the properties of arithmetical functions, including the Möbius function, Euler totient function, and their relationships and products.	An	C	Internal Exam/Assignment /Seminar/ Viva / End Sem Exam
CO2	Apply Dirichlet multiplication and inversion formulas to solve problems involving arithmetical functions, including the Mangoldt function and Liouville's function.	Ap	P	Internal Exam/Assignment /Seminar/ Viva/ End Sem Exam
CO3	Evaluate and create asymptotic formulas and theorems related to the distribution of prime numbers and quadratic residues, utilizing tools such as Chebyshev's functions and the quadratic reciprocity law.	C	F	Internal Exam/Assignment /Seminar/ Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook		1. Introduction to Analytic Number Theory, Tom M. Apostol, Springer International Student Edition, Narosa Publishing House, New Delhi, 1990 2. A course in Number Theory and Cryptography, second Edition, Neal Koblitz Springer, 1991		
Module	Unit	Content	Hrs (48+ 12)	Marks Ext: 70
I		Arithmetical Functions and their properties	18	Min.15
		Arithmetical Functions and Dirichlet Multiplication		
	1	Section 2.1- Introduction		
	2	Section 2.2- The Mobius function $\mu(n)$		
	3	Section 2.3- The Euler totient function $\phi(n)$		
	4	Section 2.4- A relation connecting μ and ϕ		
	5	Section 2.5- A product formula for $\phi(n)$		
	6	Section 2.6- The Dirichlet product of arithmetical functions		
	7	Section 2.7- Dirichlet inverses and Mobius inversion formula		
	8	Section 2.8- The Mangoldt function $\Lambda(n)$		
	9	Section 2.9- Multiplicative functions		
	10	Section 2.10- Multiplicative functions and Dirichlet Multiplication		
	11	Section 2.11- Inverse of a completely multiplicative function		
	12	Section 2.12- Liouville's function $\lambda(n)$		
II	13	Section 2.13- The divisor functions $\sigma_a(n)$	10	Min.15
	14	Section 2.14- Generalized Convolutions		
		Averages of Arithmetical Functions		
	15	Section 3.1- Introduction		
	16	Section 3.2--The big oh notation. Asymptotic equality of functions		
	17	Section 3.3- Euler's Summation formula		
	18	Section 3.4- Some elementary asymptotic formulas		
III	19	Section 3.10- The Partial sums of a Dirichlet product	10	Min.15
	20	Section 3.11- Applications of $\mu(n)$ and $\Lambda(n)$		
	21	Section 3.12- Another identity for the partial sums of a Dirichlet product		
		Some Elementary Theorems on the Distribution of Prime Numbers		
	22	Section 4.1- Introduction		
IV	23	Section 4.2- Chebyshev's functions $\psi(x)$ and $\vartheta(x)$	10	Min.15
	24	Section 4.3- Relations connecting $\vartheta(x)$ and $\pi(x)$		
	25	Section 4.4- Some equivalent forms of the prime number theorem		
	26	Section 4.5- Inequalities for $\pi(n)$ and p_n		
IV		Quadratic Residues and the Quadratic Reciprocity Law	10	Min.15
	27	Section 9.1- Quadratic residues		
	28	Section 9.2- Legendre's symbol and its properties		
	29	Section 9.3- Evaluation of $(-1 p)$ and $(2 p)$		

	30	Section 9.4- Gauss' lemma		
	31	Section 9.5- The quadratic reciprocity law		
	32	Section 9.6- Applications of the reciprocity law		
V	Open Ended: Cryptography		12	
	Chapter III			
	1: Some simple cryptosystems -3 hrs 2: Enciphering Matrices-4hrs			
	Chapter IV			
	1: The idea of public key cryptography -3 hrs 2: RSA-2 hrs			

References

1. A. Beutel spacher: Cryptology; Mathematical Association of America (Incorporated); 1994
2. H. Davenport: The higher arithmetic(6th Edn.); Cambridge Univ.Press;
3. G. H. Hardy and E.M. Wright: Introduction to the theory of numbers; Oxford International Edn; 1985
4. A. Hurwitz & N. Kritiko: Lectures on Number Theory; Springer Verlag ,Universi text;1986
5. T. Koshy: Elementary Number Theory with Applications; Harcourt / Academic Press;2002
6. D. Redmond: Number Theory; Monographs & Texts in Mathematics No: 220; Mar cel Dekker Inc.; 1994
7. P. Ribenboim: The little book of Big Primes; Springer-Verlag, New York; 1991
8. K.H. Rosen: Elementary Number Theory and its applications(3rd Edn.); Addison WesleyPub Co.; 1993
9. W. Stallings: Cryptography and Network Security-Principles and Practices; PHI; 2004
10. D.R. Stinson: Cryptography- Theory and Practice(2nd Edn.); Chapman & Hall / CRC (214. Simon Sing: The Code Book The Fourth Estate London); 1999
11. J. Stopple: A Primer of Analytic Number Theory-From Pythagoras to Riemann; Cambridge Univ Press; 2003
12. S.Y. Yan: Number Theory for Computing(2nd Edn.); Springer-Verlag; 2002

***70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	2	1	1	3	0	3	1	3	0	2
CO 2	2	3	2	1	3	0	3	2	3	0	3
CO 3	3	2	3	2	3	0	3	1	3	0	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓